

Biomedical Engineering Laboratory I

Lab 3 Microfluidics

OBJECTIVES

1. Become familiar with basic soft lithography fabrication techniques that are used to design and produce microfluidic chips;
2. Create microfluidic channels of various morphologies and test laminar flow in a microfluidic chip;
3. Characterize a simple model of the microvascular system by tracking movements of microbeads flowing through a microfluidic chip;
4. Analyze flow data using proper statistical methods.

BACKGROUND

The microvasculature, i.e., the portion of the circulatory system composed of the smallest vessels, such as the capillaries, arterioles, and venules, consists of a series of vessels and networks with extensive

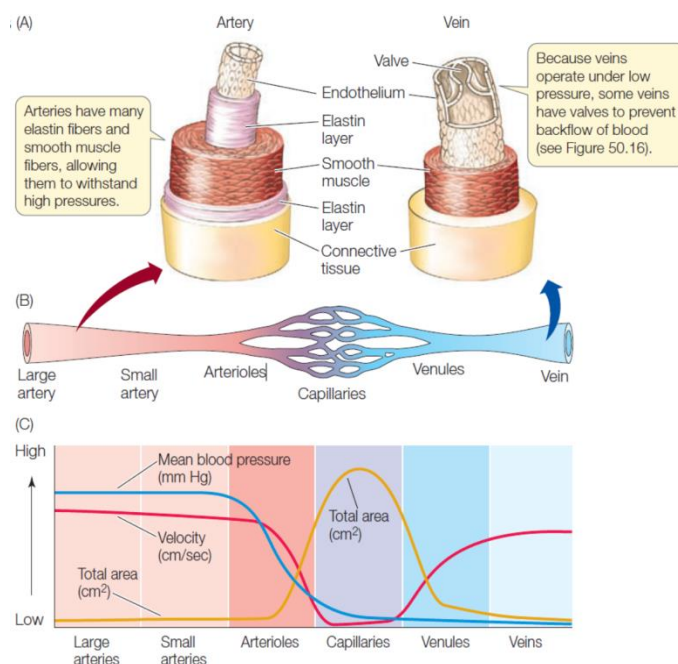


Figure 1. Branching of a typical microvasculature showing change in blood pressure, total area, and velocity.

redundant connections (Figure 1). These superfluous connections are necessary to provide a high surface area for perfusion to the various tissues in the body. It is often very difficult to model the microvasculature to determine the hemodynamic characteristics of these systems. However, the advent of soft lithography techniques has facilitated the creation of microfluidic flow systems to emulate flow conditions in the microvasculature. Consequently, it is possible to study microfluidics *in vitro* and

to simulate microvascular flow.

The creation of microfluidic systems using soft lithography is also very beneficial in the creation of low cost, rapid diagnostic devices. For example, the Sia group has created [devices](#) that can be used with a smart phone to detect sexually transmitted diseases including HIV and syphilis (Figure 2). The device performs enzyme-linked immunosorbent assays (ELISAs), which is normally an expensive and time-consuming test, on small volumes of blood. In this laboratory procedure, you will use lithography to fabricate channels that will subsequently be used for examination of microfluidic flow.



Figure 2. Sia lab's microfluidic smartphone dongle for low-cost diagnostics.

Soft Lithography

Soft lithography is the process of using light-cured polymeric materials to create micro- or nano-scale structures. The principles that govern the formation of microstructures using lithography are based on techniques developed for microfabrication of integrated circuits. However, rather than patterning semiconductor elements on silicon wafers, soft lithography for microfluidic applications entails the creation of relief structures on a wafer then creating an elastomeric mold from these structures that is subsequently bonded to a surface to create microfluidic channels. The basic steps in soft lithography are listed below and displayed in Figure 3.

1. Silicon (Si) wafer forms the substrate.
2. A photoresist (PR) layer is deposited on the Si wafer. Spin coating is often used to ensure a uniform thickness across the entire wafer. The wafer is subsequently baked to remove any excess moisture and to cure the PR.

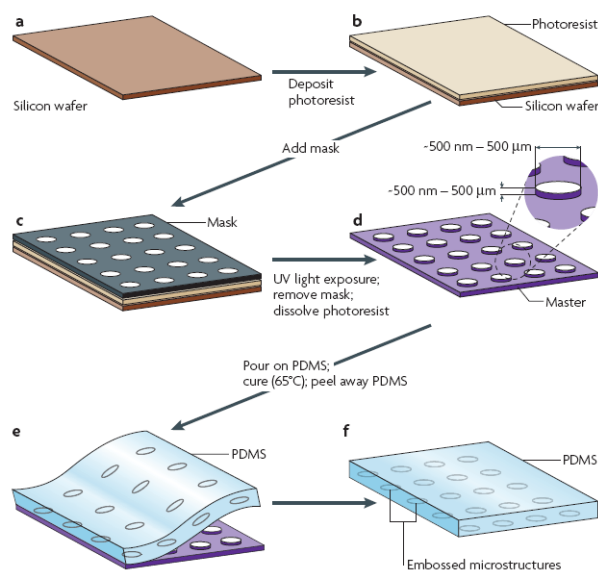


Figure 3. Basic steps in soft lithography fabrication of a PDMS mold [3]

3. A photo mask, usually a film or a glass substrate with a patterned layer of UV reflecting chromium, is overlaid on the PR.
4. The wafer-PR-mask assembly is exposed to UV light. In a negative PR, e.g., SU-8, the UV light crosslinks the exposed areas. The mask is removed and the wafer-polymer is placed in a solvent for development. The cross-linked portions of the PR remain after development, while the un-exposed regions are dissolved, leaving behind a mold of cross-linked PR on a Si wafer.
5. A stamp or mold is created by pouring a liquid elastomer, e.g., PDMS, over the master. The PDMS is solidified by heating.
6. The PDMS stamped is gently peeled from the master, creating a flexible, three dimensional structures that can be used for subsequent experimentation and testing.

The resultant PDMS stamp can be made hydrophilic by plasma preparation, which will incorporate oxygen (O) atoms into the PDMS. Additionally, the plasma preparation makes it possible to bond PDMS to a glass or silicon substrate, which are often used for microfluidics studies (plasma preparation promotes the formation of covalent Si-O-Si bonds between the PDMS stamp and the Si-based backing [1]).

Soft lithography microfabrication is relatively cheap and readily available to a variety of users; it can be used to form three dimensional micro-structures; and the techniques may be applied to a variety of materials [1-3]. Accordingly, it has become a very popular technique for investigators examining microfluidics or developing small scale assays.

Microfluidic Devices

Microfluidics entails the transport of small (micron) volumes of sample across channels on microfluidic chips. Such samples can be used to perform small scale assays, electrophoresis, and particle sorting and counting. Microfluidics is particularly useful in bio- and immunoassays, where the small sample sizes can promote high throughput screenings and the simultaneous assessment of several agents.

Fluid flow is commonly achieved in a micro-channel by generating a pressure drop across the length of a microchannel. This pressure-driven flow can be quantified by applying Ohm's law to the microchannel:

$$\Delta P = QR \quad (1)$$

Where ΔP is the pressure drop across the channel (Pa), Q is the volumetric flow rate (m^3/s) and R is the resistance to flow in the channel ($\text{Pa} \cdot \text{s}/\text{m}^3$) and is dependent on channel geometry and fluid properties:

$$R = \frac{8\mu L}{\pi r^4} \quad \text{or} \quad R = \frac{12\mu L}{wh^3} \quad (2)$$

Where μ is the fluid viscosity, L is the channel length, r is the radius of the circular channel, w and h are the width and height of rectangular channel respectively. This is the resistance to flow in a tube and is analogous to electrical resistance to current flow. In fact, many of the rules that govern networks of tubes in parallel and series are similar to those occur in electrical resistive circuits. An advantage of using microchannels for assays is that laminar flow can be readily achieved. Laminar flow is predicted using the Reynolds number:

$$Re = \frac{vlp}{\mu} \quad l = \frac{4A}{P} = \frac{4wh}{2(w+h)} \approx 2h \quad (3)$$

Where v is the fluid velocity (m/s), l is the characteristic (cross-sectional) dimension (m) of the channel, p is the fluid density (kg/m^3). When the dimensionless $Re > 2000$, turbulent flow occurs; however, for microchannels containing aqueous solutions, the small l often yields $Re \ll 1$. Thus laminar flow is common in single or multiple streams in microfluidic channels. Parallel streams are particularly interesting because mixing only occurs at the interface between multiple fluids. The streams will otherwise remain in individual laminar conditions, facilitating examination of multiple samples without unwanted mixing. A more detailed treatise on this and other characteristics of microfluidic flow can be found in [2].

In this laboratory, you will use soft lithography to create single and multi-channel PDMS stamps. The resultant stamps will be bonded to glass slides and various media will be flown through the channel. Microscopic images of the flow behavior will be obtained throughout the experiments. The images will be subsequently analyzed using ImageJ (or an image analysis software package of your choosing) to determine how channel geometry may affect flow rates and the onset of turbulence in a microchannel.

MATERIALS AND METHODS

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|-----------------|-----------------------|
| ♦ Silicon Wafer | ♦ Knives and Scrapers |
| ♦ Spinner | ♦ 0.75 mm Punch |

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| ♦ Hot Plate (95 °C) | ♦ Plasma Cleaner |
| ♦ UV Lamp | ♦ Microscope Slide |
| ♦ Film Mask | ♦ Scotch tape |
| ♦ SU-8 2025 Photoresist | ♦ Bake Oven or Hot Plate (125 °C) |
| ♦ SU-8 Developer | ♦ Steel and Silicon tubing |
| ♦ Isopropanol | ♦ 1 mL Syringe |
| ♦ Wafer Tweezers | ♦ Needles #6 |
| ♦ PDMS (Dow Corning, Sylgarg 184) | ♦ Scissors |
| ♦ Plastic Cup | ♦ Inverted Microscope |
| ♦ Electronic Balance | ♦ Fluorescent Dye |
| ♦ Glass Stirrer | ♦ Deionized Water |
| ♦ Petri Dish (10 cm) | ♦ Microbeads Suspension (9.0-9.9 μm diameter) |
| ♦ Silicon Wafer with PR micro-structure | ♦ Beaker |
| ♦ Vacuum Desiccator with Pump | ♦ Syringe Pump |
| ♦ Bake Oven (80 °C) | |

LABORATORY PROCEDURES

Procedure 1: Lithographic Fabrication of Microfluidic Channels

An advantage of soft lithography is that the techniques involved in the fabrication are relatively inexpensive and can be readily learned by a novice user. In this procedure you will create a PDMS stamp using previously discussed soft lithography techniques. There are several delicate elements to this procedure, thus you will be extensively guided by the instructor and TA.

♦ Master Fabrication (Silicon wafer with PR micro-structure).

- 1-1. Clean the silicon wafer and center it on the chuck of the spinner. Activate the vacuum to secure the silicon wafer to the chuck.
- 1-2. Deposit 2 mL of SU-8 2025 photoresist (PR) onto the center of the wafer.
- 1-3. Press RUN to activate the spinner. The spinner is set to a specific speed to yield a desired PR thickness. In this procedure the spinner is preprogrammed to spin at 500 rpm for 10 s and then 3000 rpm for 30 s to deposit an even layer of PR $\sim 25 \mu\text{m}$ thick.
- 1-4. Turn off the vacuum and place the PR-coated silicon wafer on a hot plate and bake it at **95 °C for 7 minutes**. The purpose of this step is to drive off excess solvent in the PR.
- 1-5. After the 7-minute bake, remove the silicon wafer from the hot plate and let it cool.

- 1-6. Center the silicon wafer under the UV lamp. Place the film mask and a thick glass plate on top of the mask to ensure good contact between it and the underlying silicon wafer.
- 1-7. Close the door of the **UV lamp** and press ENTER to begin exposure. The UV lamp is preprogrammed to run for **9 s**.
- 1-8. Remove the mask from the silicon wafer. Transfer the wafer onto a hot plate. Bake it at **95 °C for 7 minutes**. Set it aside to cool down.
- 1-9. Transfer the silicon wafer into a container of SU-8 developer inside the chemical fume hood. Gently agitate the container for **5 minutes**. During this time the uncrosslinked PR is dissolved away while the crosslinked (UV-exposed) PR remains on the silicon wafer.
- 1-10. Remove the silicon wafer from the dish and rinse it thoroughly with isopropanol. **If the features are not fully developed, you will see white residues on the wafer.** Spray the slide with SU-8 developer and rinse it with isopropanol again.
- 1-11. Dry the silicon wafer with nitrogen. At this point you should see clear features on the wafer. Then bake the wafer to harden the PR by placing it into a **95 °C bake oven for 2 hours**.
- ♦ *PDMS replica molding (PDMS stamp from the master)*
- 1-12. Rinse the fabricated master (i.e., the silicon wafer with UV-cured PR on it) with DI water then isopropanol to remove any dust and residues; dry with nitrogen, **be careful not to damage the PR micro-structure**.
- 1-13. Pour 23 g of PDMS base into a plastic cup. Tare the balance. Add 2.3 g of PDMS curing agent to the PDMS base. Using a glass stirrer, vigorously mix the two components thoroughly until the mixture looks very bubbly (**gives a milky white color**).
- 1-14. Degas the mixture by placing the plastic cup with the mixture in a desiccator and vacuuming it. That will take 15 - 20 minutes. After degassing, take out the plastic cup from the desiccator and remove residual bubbles on the top using a rubber bulb.
- 1-15. Pour the degassed PDMS into the dish with silicon wafer. Again, place it in a desiccator to remove all the bubbles. This time 10 minutes will be enough.
- 1-16. Transfer the Petri dish to an oven at 80 °C for 1 hour to fully cure the PDMS.

Procedure 2: Assembling a Microfluidic Device

Pouring the liquid PDMS over the master creates grooved channels in the solidified PDMS. Microchannels can be created from these grooves by bonding the PDMS to a Si-based backing. The backing and PDMS are first treated in a plasma cleaner, which introduces O atoms into both materials. After plasma treatment, the PDMS and backing pressed together, creating tight Si-O-Si bonds between the stamp and the backing. In this procedure you will assemble a microfluidic chip and test its patency and the quality of the seals on chip.

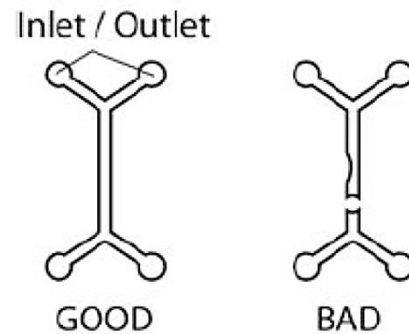


Figure 4. Illustration of correct and incorrect PDMS channels in microfluidics device.

- 2-1. Examine the microfluidic channels. Make sure you select channels that are good (e.g. no discontinuities as shown in Figure 4).
- 2-2. Carefully cut out the PDMS microchannel section using a knife and a scraper. Be sure not to cut too close to the channel!
- 2-3. Using a 0.75 mm-diameter punch, practice punching holes by perforating a piece of blank PDMS. Once comfortable with the punching technique, poke through the designated inlet and outlet areas of the PDMS. The holes should be free of cracks when observed by the naked eye (ensures that the channels will not leak).
- 2-4. Remove any dust from the PDMS using scotch tape. Leave the tape on the PDMS.
- 2-5. Clean the microscope slide with an ethanol wipe. Avoid leaving any lint or dust on the slide surface. Residual particulates may interfere with the PDMS channel bonding.
- 2-6. Remove the tape from the PDMS channel. Place it in the plasma cleaner with the channel side facing up. Place the clean microscope slide next to it. Make sure both the PDMS and glass slide are NOT covered.
- 2-7. Close the door of the plasma cleaner; turn on the vacuum pump.
- 2-8. Turn on the plasma cleaner.
- 2-9. The plasma should become a purple/pink color. This indicates an oxygen-rich plasma is generated, which oxidizes the glass and PDMS in order to form an irreversible seal.
- 2-10. The plasma cleaner is preset to be turn off automatically after 1 minute.
- 2-11. **After the voltage in the plasma cleaner decreasing down to safe range**, open the door. Remove

the PDMS and glass slide from the plasma cleaner. Immediately bond the PDMS channel to the glass slide - ***channel side facing the glass!*** Leave the assembly on a hot plate at 125 °C for 30 minutes to enhance the bonding.

- 2-12. Connect steel tubing and silicon tubing by inserting the steel one into the silicon then insert the other end of the steel tubing into one inlet/outlet on your microchannel.
- 2-13. Place the microchannel assembly on the microscope stage and obtain microscopic images of various positions along the channel when necessary. Do not forget the scale bar!
- 2-14. Place the inlet(s) into a reservoir of deionized water.
- 2-15. You will apply suction to pull fluid through the channel. Insert a #6 needle into the open end of the outlet tubing, and connect the needle to a 1 mL syringe. Retract the syringe plunger and examine the PDMS-slide assembly for leaks. Also ensure that there is flow through the channel by observing water accumulation in the syringe.
- 2-16. In Procedure 4, you will take movies (image sequences) and calculate the microbeads' velocities using any software of your choosing (e.g. ImageJ, MATLAB). The movies captured in 'Ocular' or 'Capture' can be saved in several different formats. Try all the available formats in your software analyzing the images to find the best format for you. This step is necessary for saving your time in Procedure 4.

Procedure 3: Testing Laminar Flow in a Microfluidic Chip

An advantage of studying flow in microfluidic channels is that, due to the small channel cross-sectional dimension (on the order of microns) the Reynolds number is usually quite small, $Re < 1$. Accordingly, laminar flow may be maintained, even with multiple streams of fluid. This facilitates simultaneous examination and transport of several different samples, which in turn allows for high throughput assays on a microfluidic chip. In this laboratory procedure, you will examine laminar flow in multiple streams traversing a microchannel.

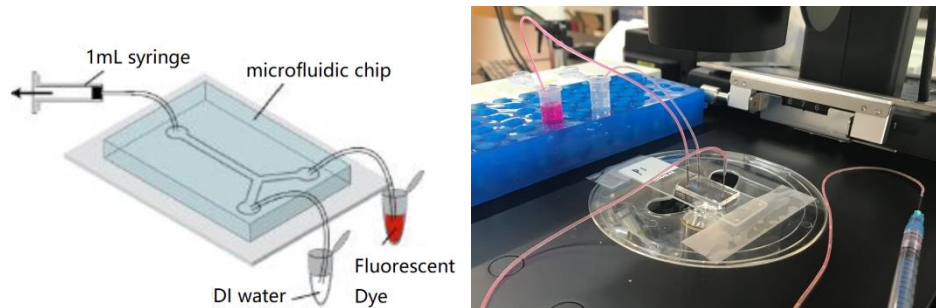


Figure 5. Schematic and photographic image of setup for testing laminar flow.

- 3-1. Carefully cut out and peel off the PDMS microchannel sections of Figure 5 & 6 which were fabricated in Procedure 1.
- 3-2. For the microchannel in Figure 5, repeat steps **2-2 through 2-11**. Set up the assembled chip as depicted in Figure 5.
- 3-3. Obtain a vial of dye and a vial of deionized water.
- 3-4. Fill a 1 mL syringe with 500 μL of DI water. Remove any air bubbles by inverting the syringe (plunger side down) and depressing the plunger. Connect the syringe to the needle that is attached to the outlet tubing of the microfluidic chip.
- 3-5. ***To ensure that there are no bubbles in the chip, you should lightly depress the syringe until you see drops of water exiting the end of the tubes before dipping the tubes into the dye and DI water.***
- 3-6. Using a 4x objective, focus on the middle section of the chip. When the position is established, secure the chip on the microscope stage with tape.
- 3-7. Gently pull the syringe to establish flow. Observe and take fluorescent images of the flow in the middle section of the channel. Do the streams mix? Why or why not?
- 3-8. Devise a method to quantify flow and calculate the Reynolds number for the flow system (state appropriate assumptions). Based on this calculation, do you expect turbulent or laminar flow in this microfluidic system? Do you expect laminar or turbulent flow in the microvasculature? Consider different vessel sizes and physiological variables. Be sure to cite any references.

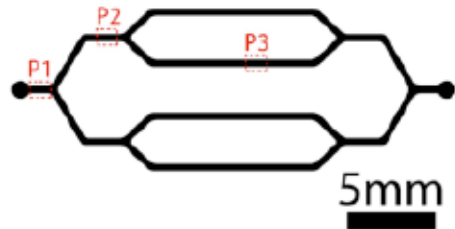
Procedure 4: Modeling a Microvascular Network on a Microfluidic Chip

The flow behavior that occurs at the anastomoses between arterioles, capillaries, and venules yields important information about how tissues are perfused in the microvasculature. Microfluidics facilitates in

vitro modeling of the microvasculature by allowing for the formation of complex and organized microchannel networks. In this laboratory procedure, you will examine flow in a microfluidic network that is designed to simulate the microvasculature.

- 4-1. For the microchannel in Figure 6 you cut out in step 3-1, repeat steps **2-2 through 2-11**.
- 4-2. Insert separate tubing at the channels' inlet and outlet. Insert a #6 needle into the open end of the outlet tubing.

- 4-3. Fill a 1 mL syringe with DI water. Eliminate any air bubbles from the syringe. Any bubbles in your system will compromise your experiment.



- 4-4. Connect the syringe to the needle in the outlet tubing. Gently depress the plunger so that drops of water are coming out of the inlet tubing. Push out any air bubbles in your chip.

Figure 6. A microfluidic channel model of a microvascular network. Channels in series add like resistors in series, channels in parallel behave like resistances in parallel.

- 4-5. Position the microfluidic chip on the microscope stage and focus so that you obtain a 10X image at the first position, **P1** in Figure 6, of the network. Use scotch tape to secure the assembled device to the microscope stage when necessary.
- 4-6. Load the syringe with microbeads solution on the syringe pump as depicted in Figure 7. **In order to avoiding the aggregation of microbeads, you should frequently knock the syringe.**

- 4-7. Configure the pump Infuse in **4 $\mu\text{L}/\text{min}$** flow rate with Air-Tite 1 mL syringe.

- 4-8. Press “” to start the flow. Observe beads flowing across your microchannel.

- 4-9. Allow the flow to equilibrate for approximate 10 seconds. Start recording a 10 s video on the camera.

Note down the frame rate so that you can calculate the time lapse between consecutive images. If the beads

are flowing too fast, i.e., they cannot be seen in the video, decrease the pump flow rate such that you can measure bead velocities from your images (be sure to recover the video files before leaving the lab for the day.)

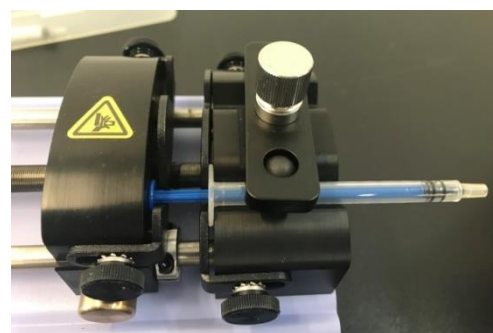


Figure 7. Example of syringe positioning in the pump.

- 4-10. Repeat Steps **4-9 through 4-11** for positions **P2** and **P3** of the channel.
- 4-11. Save the videos in the format(s) you selected after Procedure 2. Read the videos frame-by-frame in your software. Using the coordinates of the particles you are tracking and their position in consecutive images, along with the known capture rate (the frame rate of the camera), you should be able to determine linear flow velocity. **Track at least 10 particles per channel ($n \geq 10$) to get statistically relevant data.**
- 4-12. Plot velocity histograms for beads examined at positions P1, P2, and P3.
- 4-13. Plot the discrete cumulative probability distributions for the beads examined at positions P1, P2, and P3. If possible, display the distributions for all three channels on the same axes. Be sure to comment on shifts in the plots and whether this agrees with expectations.
- 4-14. Calculate the mean, median, and standard deviation for each of the 3 sets of measurements.
- 4-15. Perform a one-way analysis of variance (ANOVA) to test whether the differences in average velocities in the three channel regions are statistically significant. Use appropriate post-hoc testing. On average, for which position within the channel, P1, P2, or P3, is the bead velocity highest? Where is it the lowest? Does the difference, or lack thereof, align with your expectations? Why or why not (Hint: Use Eqns. 1 and 2)?

Laboratory Report Guidelines

Be sure to address the following in your laboratory report:

- Answer/discuss all the italicized and underlined portions and include all the requisite plots to display your data.
- Provide sample images of the various microfluidic channels and flow conditions developed during the lab (including scale bars).
- Provide sample calculations and images for determining flow velocity.
- **Ethics Question:** Elaborate on the limitations and assumptions in your experiment. Discuss what your results mean in the context of the field of microfluidics and what would be the implications of a more complex microchannel network. Do you feel that microfluidics can be used to approximate the hemodynamic conditions in a microvascular network? Why or why not? How could you change or improve the model using a microfluidic chip?

References/Relevant Reading

1. Xia, Y. and Whitesides G., Soft Lithography. Annu. Rev. Mater. Sci., 1998, 28: 153-184.
2. Sia, S. and Whitesides G., Microfluidic devices fabricated in poly(dimethylsiloxane) for biological studies. Electrophoresis, 2003, 24: 3563-3576.
3. Weibel D. B., DiLuzo W.R. and Whitesides G.M., Microfabrication meets microbiology, Nat. Rev. Microbiol., 2007, 5(3): 208-218.